MULTIPLE EMBEDDING STEGANOGRAPHY FOR RGB IMAGE

A THESIS SUBMITTED TO
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BY

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APPROVAL PAGE

Approval of the School of Computing of Telkom University

I certify that this thesis satisfies all the requirements as a thesis for the degree of Master Forensic Science.

Date Jan 09 , 2026 (*the date can be set manully)

(Dr. Farah Afianti)

Head of Master Forensic Science

This is to certify that we have read this thesis and that in our opinion it is fully adequate, in scope and quality, as a thesis for the degree of Master of Forensic Science.

Date Jan 09 , 2026

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(Jury's name) (jury's member) :

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SELF DECLARATION AGAINST PLAGIARISM

I hereby declare that all information in this document has been obtained and presented in accordance with academic rules and ethical conduct. I also declare that, as required by these rules and conduct, I have fully cited and referenced all material and results that are not original to this work.

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Signature: __________________________

Month/Date/Year Jan 09 , 2026
Name, last name of the Co-Supervisor: Co-Supervisor Name

Signature: __________________________

ABSTRACT

The demand for high-capacity digital watermarking, particularly for embedding entire images for robust copyright protection, continually pushes the boundaries of steganographic techniques. While recent methods combining Pixel Value Differencing (PVD) and edge detection in the RGB color space [3] have notably increased payload capacity, they often employ complex, sequential embedding processes where data is hidden across channels in a dependent manner. This sequential approach can introduce inter-channel dependencies and limit the ultimate payload, posing a challenge for embedding visually complex, high-resolution watermarks.

This research proposes a novel Multiple Embedding Steganography (MES) framework designed to overcome these limitations by introducing a parallel embedding architecture. Instead of a sequential process, our method treats the Red, Green, and Blue channels as three independent, parallel data carriers. The watermark image is first decomposed, and its constituent data segments are embedded simultaneously into designated bit-planes of each respective color channel. This parallel architecture maximizes the spatial utilization of the RGB color space and decouples the complex channel adjustments required by previous works. Furthermore, the embedding strength within each channel is adaptively controlled, allowing for greater data density in textured or edge regions to preserve high imperceptibility, building upon established principles of the Human Visual System (HVS).

The primary expected outcome is a substantial increase in embedding capacity, projected to surpass the 3.0 bpp ceiling of recent sequential PVD methods, thereby enabling the practical embedding of a full-size image as a watermark. This is achieved while maintaining high visual quality, with a target Peak Signal-to-Noise Ratio (PSNR) value well above the 30 dB threshold considered imperceptible to the human eye. The parallel embedding strategy is also hypothesized to distribute statistical artifacts more evenly, enhancing resilience against steganalysis. This research contributes a more robust and high-capacity steganographic framework tailored for the demanding application of image-in-image watermarking, presenting a significant advancement in balancing capacity and imperceptibility.

Keywords: Multiple Embedding, Steganography, Image Watermarking, RGB Color Space, High Capacity, Parallel Embedding, Imperceptibility, PSNR

ABSTRAK

Abstract in bahasa Indonesia.

Kata kunci:

DEDICATION

This thesis is compiled with the support of my family. I praise GOD and thanks to all who have helped, directed, and supported this work. I dedicate the work to my beloved parents ALICE and BOB.

Finally this thesis is dedicated to all informatics students in the world. I hope that this research may provide valuable contribution in Computer Science.

ACKNOWLEDGMENTS

This thesis is compiled with the effort, help, and support from both students and lecturers. I would like to express my deepest gratitude and thanks to:

1. .....

2. .....

PREFACE

In thesis writing, the most difficult part to write is Chapter 1 (Introduction/The Problem). As they say, the most difficult part of any endeavor is the starting point. This is because the first chapter is where you conceptualize your entire research. The whole research/thesis can be reflected in Chapter 1 including expected results or outcomes. For your guidelines, please read the following sample format of Chapter 1.

Bandung,

YOUR NAME

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LIST OF TERMS

Terms	Definition
Steganography	The art and science of hiding a secret message within another, non-secret message or object (the cover) to conceal the very existence of the secret message.
Cover-Image	The original, ordinary image (in this case, an RGB image) that is used as a medium to hide the secret data.
Stego-Image	The resulting image after the secret data (payload) has been successfully embedded into the cover-image.
Payload	The secret information or data that is to be hidden inside the cover-image.
Embedding	The process of inserting the payload into the cover-image using a specific steganography algorithm.
Extraction	The reverse process of embedding, where the hidden payload is retrieved from the stego-image.
RGB Image	An image that uses the RGB (Red, Green, Blue) color model. Each pixel in the image is represented by a combination of intensities of these three primary colors.
Pixel	The smallest addressable element in a digital image. Each pixel contains color and brightness information, typically as values for its color channels (e.g., R, G, and B).
LSB	*Least Significant Bit.* The rightmost bit in a binary number. LSB substitution is a common steganographic technique where the LSBs of the cover-image's pixel data are replaced with the bits of the payload.
Bit Plane	A set of bits corresponding to a specific bit position in each of the binary numbers representing the pixels. An 8-bit grayscale image, for example, can be decomposed into 8 separate bit-planes.
Imperceptibility	A crucial requirement in steganography, referring to the inability of the human visual system (HVS) to distinguish between the cover-image and the stego-image.
Capacity	The maximum amount of secret data that can be embedded into a cover-image without causing statistically significant or noticeable distortion.

Terms	Definition
PSNR	*Peak Signal-to-Noise Ratio.* A common metric used to measure the quality of the stego-image by quantifying the level of distortion compared to the cover-image. A higher PSNR value generally indicates better imperceptibility.
MSE	*Mean Squared Error.* A metric that measures the average of the squares of the differences between the pixel values of the cover-image and the stego-image. It is often used to calculate PSNR.
Steganalysis	The discipline dedicated to detecting the presence of hidden messages in media. It is the counterpart to steganography.
Cryptography	The practice of securing communication by converting a message into an unreadable format (ciphertext). Unlike steganography which hides the message's existence, cryptography obscures its content.

LIST OF NOTATIONS

Symbols	**Definition**
C	The cover-image (citra penampung)
S	The stego-image (citra hasil penyisipan)
$M \times N$	The dimensions of the image (width $\times$ height) in pixels
P	The secret message or payload to be embedded
b_k	The k-th bit of the secret message P
$C(i, j)$	The pixel value at coordinate (i, j) of the cover-image
$S(i, j)$	The pixel value at coordinate (i, j) of the stego-image
C_R, C_G, C_B	The Red, Green, and Blue channels of a cover-image
L	Number of Least Significant Bits (LSB) used for embedding
bpp	Bits Per Pixel, a measure of embedding capacity
MAX_I	Maximum possible pixel intensity value (e.g., 255 for an 8-bit image)
MSE	Mean Squared Error
$PSNR$	Peak Signal-to-Noise Ratio, in decibels (dB)
Σ	Summation operator
$\oplus$	The Exclusive OR (XOR) operator
$E(\cdot)$	The embedding function or algorithm
$D(\cdot)$	The extraction or decoding function

CHAPTER 1

INTRODUCTION

This chapter includes the following subtopics, namely: (1) Rationale; (2) Theoretical Framework; (3) Conceptual Framework/Paradigm; (4) Statement of the problem; (5) Hypothesis (Optional); (6) Assumption (Optional); (7) Scope and Delimitation; and (8) Importance of the study.

1.1 Rationale

This section have to explain: (a) the background of the study; (b) describe the problem situation considering global, national and local forces; (c) Justify the existence of the problem situation by citing statistical data and authoritative sources; and (d) Make a clinching statement that will relate the background to the proposed research problem.

1.2 Theoretical Framework

Discuss the theories and/or concepts, which are useful in conceptualizing the research.

1.3 Conceptual Framework/Paradigm

Identify and discuss the variables related to the problem, and present a schematic diagram of the paradigm of the research and discuss the relationship of the elements/variables therein.

1.4 Statement of the Problem

The general problem must be reflective of the title. It should be stated in such a way that it is not answerable by yes or no, not indicative of when and where. Rather, it should reflect between and among variables. Each sub-problem should cover mutually exclusive dimensions (no overlapping). The sub-problem should be arranged in logical order from actual to analytical following the flow in the research paradigm.

1.5 Objective and Hypotheses

First, explain the objective according the problem, then hypothesis. A hypothesis should be measurable/ desirable. It expresses expected relationship between two or more variables. It is based on the theory and/or empirical evidence. There are techniques available to measure or describe the variables. It is on a one to one correspondence with the specific problems of the study. A hypothesis in statistical form has the following characteristics: (a) it is used when the test of significance of relationships and difference of measures are involved; and (b) the level of significance if stated.

1.6 Assumption

An assumption should be based on the general and specific problems. It is stated in simple, brief, generally accepted statement.

1.7 Scope and Delimitation

Indicate the principal variables, locale, timeframe, and justification.

1.8 Significance of the Study

This section describes the contributions of the study as new knowledge, make findings more conclusive. It cites the usefulness of the study to the specific groups.

CHAPTER 2

REVIEW OF LITERATURE AND STUDIES

This chapter starts with a brief introductory paragraph concerning the researcher's exploration of related literature and studies on the research problem. It should be organized thematically to confirm to the specific problems. It should synthesize evidence from all studies reviewed to get an overall understanding of the state of the knowledge in the problem area. As much as possible, the reviewed should be limited within the last ten years. A statement showing how the related materials had assisted the researchers in the present study should be the last part.

This chapter discusses the literature studies and theories of this study. It is divided into two sections: (1) the related literatures; and (2) the related studies.

2.1 Related Literatures

This section discusses state of the art of biometrics technologies and typical process which should be conducted in biometrics such as: data acquisition; preprocessing; feature extraction; and classification.

2.2 Related Studies

This section discusses about the related studies which support in the designing of the proposed method.

CHAPTER 3

RESEARCH METHODOLOGY

This chapter of a thesis commences a brief statement and enumerating the main topics that are to be covered in it; namely; (1) Research Design; (2) Sources of Data (Locale of the Study and Population/Sampling); (3) Instrumentation and Data Collection; and (4) Tools for Data Analysis. Writing chapter 3 of a thesis requires the assistance of a statistician (in most cases). This is because it is in this chapter that the thesis writer is usually required to indicate what statistical tools he intends to use in data analysis. Here is the basic format of Chapter 3.

3.1 Research Design

The appropriate research design should be specified and described (including requirement, modeling and detailed description of system/product/method development.

3.1.1 System/product/method Implementation

Describe the place where the study was conducted and the rationale behind its choice, the environment of the system (device specification, tools specification and language, which were used in the implementation)

3.1.2 Experiment Scenario

Describe the experiment scenario (including the objective, the procedure of the experiment and the variable which will be used in the experiment).

3.1.2.1 This is a subsubsection

3.1.2.1.1 This is a subsubsubsection

3.2 Population/Sampling

Describe the population of interest and the sampling of subjects used in the study.

3.3 Instrumentation and Data Collection

Describe the instrument, what it will measure, how to interpret. Discuss how the validity and the reliability will be established. Specify the level of reliability (probability). Give details of instruction given to assistants if persons other than the researcher gather data. State qualifications of informants if used in the study.

3.4 Tools for Data Analysis

Determine and justify the statistical treatment for each sub-problem, the scales of values used and the descriptive equivalent ratings, if any.

CHAPTER 4

PRESENTATION, ANALYSIS AND INTERPRETATION OF DATA

In thesis writing, the most difficult part to defend is chapter 4 because it is in this section where you will present the results of the whole study. Here is a sample thesis format.

4.1 Presentation of Data

Present the findings of the study in the order of the specific problem as stated in the statement of the Problem. Present the data in these forms: (a) tabular; (b) textual; and (c) graphical (optional). The ZOOM LENS approach may be used for purposes of clarity in the presentation of data, i.e. general to particular, macro to micro or vice versa. **(Note: Mean of data in here is data of the experiment result, not the data for input of the system)**

4.2 Analysis of the Data

Data may be analyzed quantitatively or qualitatively depending on the level of measurement and the number of dimensions and variables of the study. Analyze in depth to give meaning to the data presented in the data presented in the table. Avoid table reading. State statistical descriptions in declarative sentences, e.g. in the studies involving:

1. Correlation
    (a) State level of correlation
    (b) State whether positive or negative
    (c) Indicate the level of significance
    (d) Make a decision
2. Differences of Measures
    (a) State the obtained statistical results
    (b) Indicate the level of significance of the difference
    (c) Make a decision
3. Interpretation of Data
    (a) Establish interconnection between and among data

(b) Check for indicators whether hypothesis/es is/are supported or not by findings.

(c) Link the present findings with the previous literature.

(d) Use parallel observations with contemporary events to give credence presented in the introduction.

(e) Draw out implications.

In thesis writing, the Chapter is simply a summary of what the researcher had done all throughout the whole research.

4.3 Summary of Findings

This describes the problem, research design, and the findings (answer to the questions raised). The recommended format is the paragraph form instead of the enumeration form. For each of the problems, present: (a) the salient findings; and (b) the results of the hypothesis tested.

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

5.1 Conclusions

These are brief, generalized statements in answer to the general and each of the specific sub-problems. These contain generalized in relation to the population. These are general inferences applicable to a wider and similar population. Flexibility is considered in making of conclusions. It is not a must to state conclusions on a one-to-one correspondence with the problems and the findings as all variables can be subsume in one paragraph. Conclusions may be used as generalizations from a micro to a macro-level or vice versa (ZOOM LENS approach). **Conclusions should be written on paragraph**.

5.2 Recommendations

They should be based on the findings and conclusion of the study. Recommendations may be specific or general or both. They may include suggestions for further studies. They should be in non-technical language, feasible, workable, flexible, doable, and adaptable. An action plan is optional.

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[5] Y. Zhou and A. Kumar. Contactless palm vein identification using multiple representations. In *2010 Fourth IEEE International Conference on Washington DC*, ,, pages 1–6. IEEE, Biometrics: Theory Applications and Systems (BTAS), September 2010.

Appendices

APPENDIX A

MISCELLANEOUS

This chapter discusses how to insert equation, figure and table on $\LaTeX$ document. All the object in this chapter just the examples to make you ease to write your awesome thesis.

A.1 Equation #examples

For simple equation, There is a equation $10 + 5x = 0$, then determine the value of x by solving that algebraic equation problem. The answer is shown by equation A.1

$$\begin{aligned} 10 + 5x &= 0 \\ 5x &= -10 \\ x &= -2 \end{aligned} \tag{A.1}$$

Besides that, for complicated equation will be explained as follow. The general equation for a 2D (N by M image) Discrete Cosine Transform is defined by equation A.2

$$F(u,v) = \sqrt{\frac{2}{N}}\sqrt{\frac{2}{M}}\sum_{i=0}^{N-1} A(i) * \cos\left(\frac{u(2i+1)\pi}{2N}\right) * \sum_{j=0}^{M-1} A(j) * \cos\left(\frac{v(2j+1)\pi}{2M}\right) * f(i,j) \tag{A.2}$$

where $A(i)$ is defined as $\begin{cases} \frac{1}{\sqrt{2}}\ ,\ for\ u\ =\ 0 \\ 1\ ,\ otherwise \end{cases}$, while $A(j)$ as $\begin{cases} \frac{1}{\sqrt{2}}\ ,\ for\ v\ =\ 0 \\ 1\ ,\ otherwise \end{cases}$

A.2 Figure #examples

This section contain how to insert figure (see figure A.1) and block diagram (see figure A.2). We recommend to use high resolution JPEG or JPG instead PNG format, it will make the compiling process to PDF faster, size of the PDF file is smaller and the quality is image still better when the PDF is print out or zooming out.

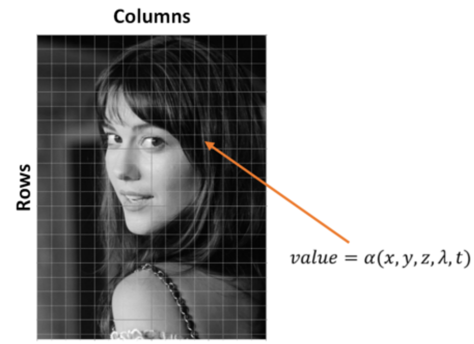

Figure A.1: Picture of mary elizabeth winstead

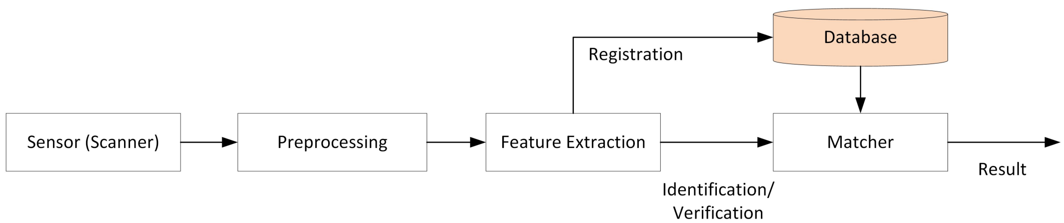

Figure A.2: Example of block diagram.

A.3 Table #examples

The regular table (see table A.1) only can be used in single page. So, for the long table which use multiple pages is recommended to use longtable (see table A.2)

1. Table
   Table A.1 will show you how to cite the reference, it can be the author or the year of the reference.

Table A.1: Example of Tables

Reference	**Paper or Journal**	
	Authors	***Full Authors***
[1]	Jain et al. [1][a]	Jain, Flynn, and Ross
[2]	Kong et al. [2]	Kong, Zhang, and Kamel
[4]	Wang et al. [4]	Wang, Yau, Suwandy, and Sung
[5]	Zhou and Kumar [5]	Zhou and Kumar

[a] : 2007.********.

2. Longtable

Table A.2: Iris flower data set #Example of longtable

Sepal length	**Sepal width**	**Petal length**	**Petal width**	**Species**
5.1	3.5	1.4	0.2	*I. setosa*
4.9	3	1.4	0.2	*I. setosa*
4.7	3.2	1.3	0.2	*I. setosa*
4.6	3.1	1.5	0.2	*I. setosa*
5	3.6	1.4	0.2	*I. setosa*
5.4	3.9	1.7	0.4	*I. setosa*
4.6	3.4	1.4	0.3	*I. setosa*
5	3.4	1.5	0.2	*I. setosa*
4.4	2.9	1.4	0.2	*I. setosa*
4.9	3.1	1.5	0.1	*I. setosa*
5.4	3.7	1.5	0.2	*I. setosa*
4.8	3.4	1.6	0.2	*I. setosa*
4.8	3	1.4	0.1	*I. setosa*
4.3	3	1.1	0.1	*I. setosa*
5.8	4	1.2	0.2	*I. setosa*
5.7	4.4	1.5	0.4	*I. setosa*
5.4	3.9	1.3	0.4	*I. setosa*
5.1	3.5	1.4	0.3	*I. setosa*
5.7	3.8	1.7	0.3	*I. setosa*
5.1	3.8	1.5	0.3	*I. setosa*
5.4	3.4	1.7	0.2	*I. setosa*
5.1	3.7	1.5	0.4	*I. setosa*
4.6	3.6	1	0.2	*I. setosa*
5.1	3.3	1.7	0.5	*I. setosa*
4.8	3.4	1.9	0.2	*I. setosa*
5	3	1.6	0.2	*I. setosa*
5	3.4	1.6	0.4	*I. setosa*
5.2	3.5	1.5	0.2	*I. setosa*
5.2	3.4	1.4	0.2	*I. setosa*
4.7	3.2	1.6	0.2	*I. setosa*
4.8	3.1	1.6	0.2	*I. setosa*
5.4	3.4	1.5	0.4	*I. setosa*
5.2	4.1	1.5	0.1	*I. setosa*
5.5	4.2	1.4	0.2	*I. setosa*

Sepal length	**Sepal width**	**Petal length**	**Petal width**	**Species**
4.9	3.1	1.5	0.2	*I. setosa*
5	3.2	1.2	0.2	*I. setosa*
5.5	3.5	1.3	0.2	*I. setosa*
4.9	3.6	1.4	0.1	*I. setosa*
4.4	3	1.3	0.2	*I. setosa*
5.1	3.4	1.5	0.2	*I. setosa*
5	3.5	1.3	0.3	*I. setosa*
4.5	2.3	1.3	0.3	*I. setosa*
4.4	3.2	1.3	0.2	*I. setosa*
5	3.5	1.6	0.6	*I. setosa*
5.1	3.8	1.9	0.4	*I. setosa*
4.8	3	1.4	0.3	*I. setosa*
5.1	3.8	1.6	0.2	*I. setosa*
4.6	3.2	1.4	0.2	*I. setosa*
5.3	3.7	1.5	0.2	*I. setosa*
5	3.3	1.4	0.2	*I. setosa*
7	3.2	4.7	1.4	*I. versicolor*
6.4	3.2	4.5	1.5	*I. versicolor*
6.9	3.1	4.9	1.5	*I. versicolor*
5.5	2.3	4	1.3	*I. versicolor*
6.5	2.8	4.6	1.5	*I. versicolor*
5.7	2.8	4.5	1.3	*I. versicolor*
6.3	3.3	4.7	1.6	*I. versicolor*
4.9	2.4	3.3	1	*I. versicolor*
6.6	2.9	4.6	1.3	*I. versicolor*
5.2	2.7	3.9	1.4	*I. versicolor*
5	2	3.5	1	*I. versicolor*
5.9	3	4.2	1.5	*I. versicolor*
6	2.2	4	1	*I. versicolor*
6.1	2.9	4.7	1.4	*I. versicolor*
5.6	2.9	3.6	1.3	*I. versicolor*
6.7	3.1	4.4	1.4	*I. versicolor*
5.6	3	4.5	1.5	*I. versicolor*
5.8	2.7	4.1	1	*I. versicolor*
6.2	2.2	4.5	1.5	*I. versicolor*
5.6	2.5	3.9	1.1	*I. versicolor*
5.9	3.2	4.8	1.8	*I. versicolor*

Sepal length	**Sepal width**	**Petal length**	**Petal width**	**Species**
6.1	2.8	4	1.3	*I. versicolor*
6.3	2.5	4.9	1.5	*I. versicolor*
6.1	2.8	4.7	1.2	*I. versicolor*
6.4	2.9	4.3	1.3	*I. versicolor*
6.6	3	4.4	1.4	*I. versicolor*
6.8	2.8	4.8	1.4	*I. versicolor*
6.7	3	5	1.7	*I. versicolor*
6	2.9	4.5	1.5	*I. versicolor*
5.7	2.6	3.5	1	*I. versicolor*
5.5	2.4	3.8	1.1	*I. versicolor*
5.5	2.4	3.7	1	*I. versicolor*
5.8	2.7	3.9	1.2	*I. versicolor*
6	2.7	5.1	1.6	*I. versicolor*
5.4	3	4.5	1.5	*I. versicolor*
6	3.4	4.5	1.6	*I. versicolor*
6.7	3.1	4.7	1.5	*I. versicolor*
6.3	2.3	4.4	1.3	*I. versicolor*
5.6	3	4.1	1.3	*I. versicolor*
5.5	2.5	4	1.3	*I. versicolor*
5.5	2.6	4.4	1.2	*I. versicolor*
6.1	3	4.6	1.4	*I. versicolor*
5.8	2.6	4	1.2	*I. versicolor*
5	2.3	3.3	1	*I. versicolor*
5.6	2.7	4.2	1.3	*I. versicolor*
5.7	3	4.2	1.2	*I. versicolor*
5.7	2.9	4.2	1.3	*I. versicolor*
6.2	2.9	4.3	1.3	*I. versicolor*
5.1	2.5	3	1.1	*I. versicolor*
5.7	2.8	4.1	1.3	*I. versicolor*
6.3	3.3	6	2.5	*I. virginica*
5.8	2.7	5.1	1.9	*I. virginica*
7.1	3	5.9	2.1	*I. virginica*
6.3	2.9	5.6	1.8	*I. virginica*
6.5	3	5.8	2.2	*I. virginica*
7.6	3	6.6	2.1	*I. virginica*
4.9	2.5	4.5	1.7	*I. virginica*
7.3	2.9	6.3	1.8	*I. virginica*

Sepal length	Sepal width	Petal length	Petal width	Species
6.7	2.5	5.8	1.8	*I. virginica*
7.2	3.6	6.1	2.5	*I. virginica*
6.5	3.2	5.1	2	*I. virginica*
6.4	2.7	5.3	1.9	*I. virginica*
6.8	3	5.5	2.1	*I. virginica*
5.7	2.5	5	2	*I. virginica*
5.8	2.8	5.1	2.4	*I. virginica*
6.4	3.2	5.3	2.3	*I. virginica*
6.5	3	5.5	1.8	*I. virginica*
7.7	3.8	6.7	2.2	*I. virginica*
7.7	2.6	6.9	2.3	*I. virginica*
6	2.2	5	1.5	*I. virginica*
6.9	3.2	5.7	2.3	*I. virginica*
5.6	2.8	4.9	2	*I. virginica*
7.7	2.8	6.7	2	*I. virginica*
6.3	2.7	4.9	1.8	*I. virginica*
6.7	3.3	5.7	2.1	*I. virginica*
7.2	3.2	6	1.8	*I. virginica*
6.2	2.8	4.8	1.8	*I. virginica*
6.1	3	4.9	1.8	*I. virginica*
6.4	2.8	5.6	2.1	*I. virginica*
7.2	3	5.8	1.6	*I. virginica*
7.4	2.8	6.1	1.9	*I. virginica*
7.9	3.8	6.4	2	*I. virginica*
6.4	2.8	5.6	2.2	*I. virginica*
6.3	2.8	5.1	1.5	*I. virginica*
6.1	2.6	5.6	1.4	*I. virginica*
7.7	3	6.1	2.3	*I. virginica*
6.3	3.4	5.6	2.4	*I. virginica*
6.4	3.1	5.5	1.8	*I. virginica*
6	3	4.8	1.8	*I. virginica*
6.9	3.1	5.4	2.1	*I. virginica*
6.7	3.1	5.6	2.4	*I. virginica*
6.9	3.1	5.1	2.3	*I. virginica*
5.8	2.7	5.1	1.9	*I. virginica*
6.8	3.2	5.9	2.3	*I. virginica*
6.7	3.3	5.7	2.5	*I. virginica*

Sepal length	**Sepal width**	**Petal length**	**Petal width**	**Species**
6.7	3	5.2	2.3	*I. virginica*
6.3	2.5	5	1.9	*I. virginica*
6.5	3	5.2	2	*I. virginica*
6.2	3.4	5.4	2.3	*I. virginica*
5.9	3	5.1	1.8	*I. virginica*

APPENDIX B

Curriculum Vitae # Example

PERSONAL INFORMATION

Name	:	Mr. Handsome
Date and Place of Birth	:	May 04, 2015 - Bandung, West Java
Address	:	...
City and Province	:	...
Postal Code	:	99999
Sex	:	Male
Nationality	:	Indonesia
Religion	:	...
Marital Status	:	Single
Language Skill 1	:	Indonesian (Mother Tongue)
Language Skill 2	:	English (Sufficient)
Language Skill 3	:	Minang (Sufficient)
Language Skill 4	:	Java (Intermediate)
Language Skill 5	:	Malay (Basic Knowledge)
Language Skill 6	:	Arabic (Basic Knowledge)
Phone 1	:	+62 851 xxxx xxxx
Phone 2	:	+62 852 xxxx xxxx
Fax	:	-
Email	:	youremail@students.telkomuniversity.ac.id

EDUCATION

Education Degree	**Institution**	**Period of Education**	**Year of Graduation**
Master	Telkom University	2013 - 2015	2015
Bachelor	...	2008 - 2012	2012
High School	...	2007 - 2008	2008
Middle School	...	2003 - 2005	2005

PUBLICATIONS

Title	Event	Year
...	...	2011
...	...	2012

TEACHING AND PROFESSIONAL EXPERIENCE

Position	Institution	Field	Period
...	...	...	2012 - 2013
...	...	...	2009 - 2013

TRAINING AND WORKSHOP

Training and Workshop	Organizer	Year
...	...	2012
...	...	2009

ACHIEVEMENT

Achievement	Event	Year
1^{th}	Winner of The Awesome Thesis 2015, Telkom University	2015